A2 · Unit 4 · WPH14 Further Mechanics, Fields & Particles

Impulse & 2D Collisions

At AS you saw that momentum is conserved when things collide in a straight line. This topic finishes the story. First we look at what a force does to momentum over time, which is called **impulse** and explains crumple zones, airbags and why you bend your knees when you land. Then we take collisions off the line and onto a table top, where balls fly off at angles, and learn how to tell whether any kinetic energy was lost on the way.

Momentum and Newton's second law

Quick recap from Forces & Momentum: the momentum of an object is its mass times its velocity. Because velocity is a vector, **momentum is a vector** too. It points the same way as the velocity, and in a calculation anything moving the "wrong" way gets a minus sign.

Momentum

FORMULA SHEET

$$p = mv$$

p: momentum / kg m s^{-1} m: mass / kg v: velocity / m s^{-1}

You already know $F = ma$. Newton actually wrote his second law in terms of momentum, and that version turns out to be the more useful one for impacts and collisions. Let's see that the two say the same thing.

From F = ma to F = Δp/Δt

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Start with Newton's second law for a constant resultant force F acting on a mass m:

$$F = ma$$

2. The force is constant, so the acceleration is constant. It is the change in velocity divided by the time taken, where u is the velocity before and v the velocity after:

$$F = m\,\frac{v - u}{\Delta t}$$

3 Multiply out the bracket. The mass is constant, so it can go inside:

$$F = \frac{mv - mu}{\Delta t}$$

mv is the final momentum and mu the initial momentum, so the top line is the change in momentum Δp.

4 So the resultant force equals the rate of change of momentum:

$$\boxed{F = \frac{\Delta p}{\Delta t}}$$

We assumed the mass stays the same. $F = \Delta p / \Delta t$ still works when it doesn't, which is why Newton's own version is the more general one.

DEFINITION

Newton's second law (momentum form)

The **resultant force** acting on an object is equal to the **rate of change of its momentum**, and acts in the direction of the change in momentum.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Rocket propulsion: F = Δp/Δt for the exhaust

A rocket in space has nothing to push against, so how does it accelerate? It throws mass backwards. If the engines eject a mass Δm of hot gas every Δt, at speed u relative to the rocket, the gas gains momentum $\Delta m\, u$ every Δt. That needs a force on the gas of

$$F = \frac{\Delta p}{\Delta t} = \frac{\Delta m}{\Delta t}\, u$$

By Newton's third law the gas pushes the rocket forwards with the same size force: the **thrust**. The first stage of a Saturn V burned about 13 000 kg of propellant per second with an exhaust speed of about 2600 m s^{-1}, giving $13\,000 \times 2600 \approx 3.4 \times 10^{7}$ N. As the fuel is used up the rocket's mass falls, so the same thrust gives a bigger and bigger acceleration. Here the mass of the rocket really does change, so $F = ma$ alone would not get you there.

Impulse

Rearrange $F = \Delta p/\Delta t$ and you get the most useful equation in this topic. The quantity $F\Delta t$ (force multiplied by the time it acts for) is called the **impulse**, and it equals the change in momentum it causes.

Impulse

FORMULA SHEET

$$F\Delta t = \Delta p$$

F: resultant force / N Δt: time the force acts / s Δp: change in momentum / kg m s^{-1} or N s

DEFINITION

Impulse

The impulse of a force is the product of the force and the time for which it acts, $F\Delta t$. It is equal to the **change in momentum** of the object it acts on. Impulse is a vector, measured in **N s** (which is the same as kg m s^{-1}).

In a real impact the force is not constant: it rises as the objects squash together and falls as they separate. Each thin strip under a force–time graph has area $F\,\delta t$, which is a little bit of impulse, so adding them all up gives the key result:

KEY IDEA

Impulse is the area under a force–time graph

Area under the F–t graph = impulse = change in momentum. For a constant force the area is a rectangle, $F\Delta t$. For a varying force, count squares or split the area into triangles and rectangles. The average force is the impulse divided by the contact time.

Same change in momentum, longer time, smaller force

Suppose you need to stop something. Its change in momentum is fixed (it goes from mv to zero), so the impulse needed is fixed. But impulse is force *times* time, so if you can make the stop take ten times longer, the average force is ten times smaller. That one idea explains a lot of safety design:

- **Crumple zones** fold up over about a tenth of a second instead of the car stopping almost instantly against a wall.
- **Airbags and seatbelts** stretch or squash, so your head and chest take longer to stop.
- **Cycle helmets, crash mats and trainers** have soft foam that increases the stopping time.

- **Bending your knees** when you land, or "giving" with your hands when you catch a ball, does the same job with your own body.

Sport uses the idea the other way round. When you **follow through** with a racket, bat or kick, the force acts on the ball for longer, so for the same force the ball gets a bigger impulse and leaves faster.

WORKED EXAMPLE

A tennis ball of mass 0.058 kg hits a wall at 25 m s^{-1} and rebounds along the same line at 20 m s^{-1}. It is in contact with the wall for 4.0 ms. Calculate (a) the impulse on the ball and (b) the average force the wall exerts on it.

Show solution

(a) Choose a positive direction first: **towards the wall is positive**. Then $u = +25\ \text{m s}^{-1}$ and $v = -20\ \text{m s}^{-1}$.

$$\Delta p = m(v - u) = 0.058 \times (-20 - 25) = 0.058 \times (-45) = -2.61\ \text{N s}$$

The impulse is $2.6\ \text{N s}$ directed away from the wall.

(b)

$$F = \frac{\Delta p}{\Delta t} = \frac{-2.61}{4.0 \times 10^{-3}} = -653\ \text{N}$$

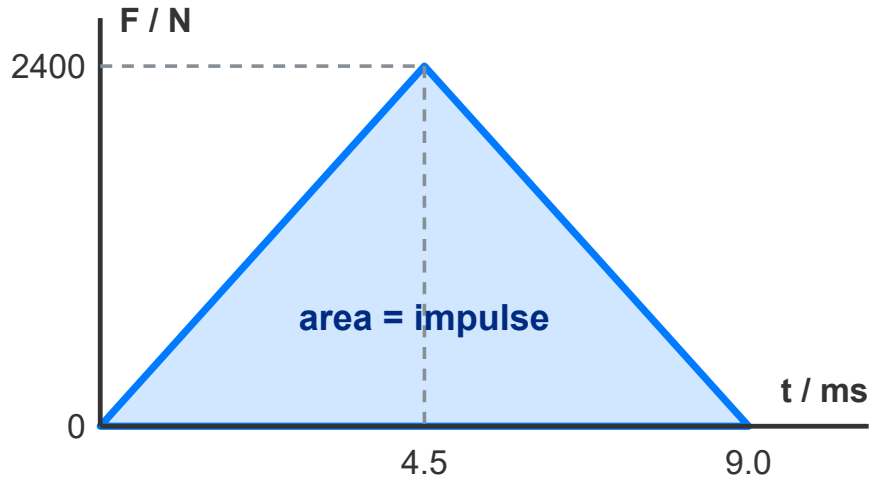
So the average force is about $650\ \text{N}$ away from the wall. That is over a thousand times the ball's weight, which is why the ball squashes visibly.

COMMON MISTAKE

When something bounces back, its velocity changes sign. Writing $\Delta p = 0.058 \times (20 - 25)$ gives $-0.29\ \text{N s}$, which is wrong by a factor of about nine. A rebound always gives a **bigger** change in momentum than just stopping, because the object has to be stopped and then sent back.

WORKED EXAMPLE

A football of mass 0.45 kg is at rest. The graph shows how the force from a boot varies during the kick. Find the speed of the ball as it leaves the boot, and the average force during the kick.



Force on the ball during the kick.

Show solution

The impulse is the area of the triangle. Convert the time to seconds first:

$$F\Delta t = \tfrac{1}{2} \times 9.0 \times 10^{-3} \times 2400 = 10.8 \text{ N s}$$

The ball starts at rest, so all of this is its final momentum:

$$v = \frac{\Delta p}{m} = \frac{10.8}{0.45} = 24 \text{ m s}^{-1}$$

The average force is the impulse spread over the whole contact time:

$$F_{\text{avg}} = \frac{10.8}{9.0 \times 10^{-3}} = 1200 \text{ N}$$

For a triangle the average force is exactly half the peak, which is a nice check.

Core Practical 9: force and change of momentum

CORE PRACTICAL

Investigate the relationship between the force exerted on an object and its change of momentum

A trolley runs along a horizontal runway, pulled by a string that passes over a pulley at the end of the bench to a hanging mass. A **force sensor** on the trolley measures the tension F in the string. Two **light gates**,

connected to a data logger, time a card of known length L on top of the trolley, giving its speeds $u = L/t_1$ and $v = L/t_2$ at the two gates, and the time Δt taken to travel between them.

1. **Compensate for friction**: raise the start end of the runway slightly until a gently pushed trolley passes through both gates at the same speed (equal card times).
2. Measure the mass m of the trolley, including the sensor and card, with a balance. Measure L with vernier calipers.
3. Release the trolley from rest. Record F, u, v and Δt. Repeat with different hanging masses, and repeat each one to get a mean.
4. For each run calculate the impulse $F\Delta t$ and the change in momentum $\Delta p = m(v - u)$.

What to plot: Δp on the y-axis against $F\Delta t$ on the x-axis. If $F\Delta t = \Delta p$, the points lie on a **straight line through the origin with a gradient of 1**. (Another version keeps F fixed and moves the second gate to change Δt. Then Δp against Δt is a straight line through the origin whose gradient is F.)

- If friction is not compensated, the resultant force is less than the tension, so every Δp comes out smaller than $F\Delta t$. The points fall below the line of gradient 1 and the best-fit line no longer passes through the origin. That is a **systematic** error.
- Keep the string parallel to the runway, so all of the tension pulls along the runway.
- Use a long card and a data logger: hand timing over these short times would swamp the result with reaction-time error.
- **Safety:** put a box with padding under the hanging masses so they don't land on feet, and a buffer at the end of the runway to stop the trolley.

Conservation of momentum in two dimensions

DEFINITION

Principle of conservation of linear momentum

In a **closed system** (one with no external resultant force acting), the **total momentum stays constant**. In a collision or explosion, the total momentum before equals the total momentum after.

Why is this true? During a collision, A pushes on B and B pushes back on A with an equal and opposite force (Newton's third law) for exactly the same time. So the impulses on A and B are equal and opposite: whatever momentum A loses, B gains. That argument works in any direction, which is the key to 2D problems.

On a snooker table or an air-hockey table, objects fly off at angles. Momentum is a vector, so we deal with it in one of two ways:

- **Resolve into components.** Pick x and y directions at right angles. Momentum is conserved separately in each direction: total p_x before = total p_x after, and total p_y before = total p_y after. That gives two equations, so you can find two unknowns (often a speed and an angle).

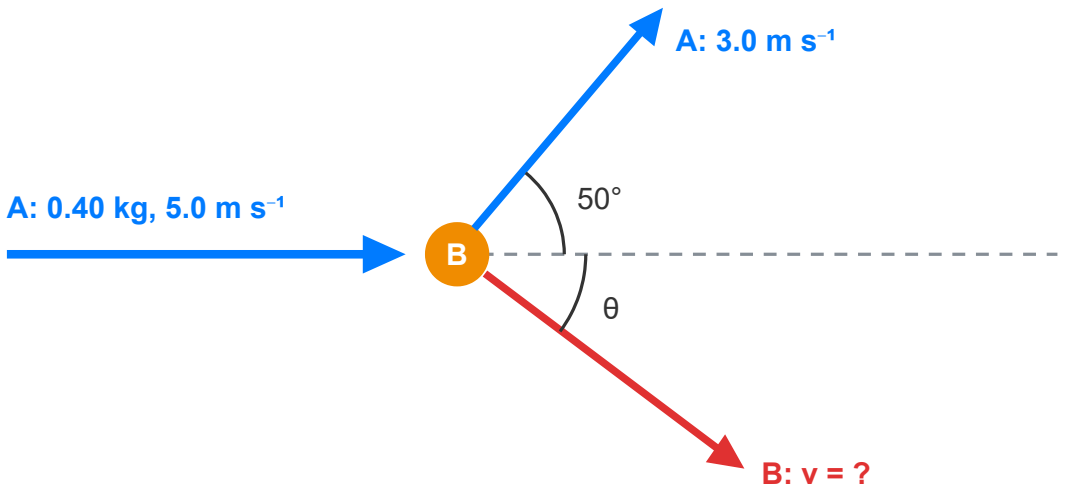
- **Draw a vector diagram.** Draw the momentum vectors after the collision tip-to-tail. Their resultant must be the same arrow as the total momentum before.

EXAM TIP

Lay the x-axis along the direction of the incoming object. Then the total p_y before is zero, so the y-components after must cancel: if one object goes off above the line, the other **must** go off below it.

WORKED EXAMPLE

Puck A (mass 0.40 kg) slides at 5.0 m s^{-1} and hits puck B (mass 0.60 kg), which is at rest. Afterwards A moves at 3.0 m s^{-1} at 50° to its original direction. Find (a) the speed and direction of B and (b) whether the collision is elastic.



Take x along A's original direction and y upwards.

Show solution

x-direction. Before: $0.40 \times 5.0 = 2.00$ kg m s^{-1}. After, A contributes $0.40 \times 3.0 \cos 50^\circ = 0.771$ kg m s^{-1}, so

$$0.60\,v_{Bx} = 2.00 - 0.771 = 1.229 \quad \Rightarrow \quad v_{Bx} = 2.05 \text{ m s}^{-1}$$

y-direction. Before: zero. After, A contributes $+0.40 \times 3.0 \sin 50^\circ = +0.919$ kg m s^{-1}, so B must have -0.919:

$$0.60\,v_{By} = -0.919 \quad \Rightarrow \quad v_{By} = -1.53 \text{ m s}^{-1}$$

Combine the components with Pythagoras and tan:

$$v_B = \sqrt{2.05^2 + 1.53^2} = 2.56 \text{ m s}^{-1}, \qquad \theta = \tan^{-1}\left(\frac{1.53}{2.05}\right) = 36.8^\circ$$

So B moves at 2.6 m s^{-1} at 37° below A's original direction.

(b) Kinetic energy is a scalar, so just add the values for each object, using the full speeds:

$$E_{\text{before}} = \tfrac{1}{2}(0.40)(5.0)^2 = 5.0 \text{ J}$$

$$E_{\text{after}} = \tfrac{1}{2}(0.40)(3.0)^2 + \tfrac{1}{2}(0.60)(2.56)^2 = 1.80 + 1.96 = 3.8 \text{ J}$$

Kinetic energy has fallen by about 1.2 J (a quarter of it), so the collision is **inelastic**.

Explosions in two dimensions

An explosion is a collision run backwards. If the object starts at rest, the total momentum before is zero, so the momenta of all the pieces afterwards must add up to zero. With two pieces, they fly apart in exactly opposite directions with equal-sized momenta (that's recoil). With three pieces, the three momentum vectors drawn tip-to-tail must form a **closed triangle**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

The 90° rule for an elastic glancing collision between equal masses

Snooker players know that when the cue ball clips a stationary ball, the two balls head off at right angles. Here is why, using nothing but the two conservation laws.

Why the paths are at 90° NOT PART OF THE EDEXCEL SPEC

1. Ball A (velocity $\vec{u}$) hits ball B at rest. Both have mass m. Afterwards they have velocities $\vec{v}_A$ and $\vec{v}_B$. Momentum is conserved (as vectors), and the m cancels:

$$m\vec{u} = m\vec{v}_A + m\vec{v}_B \quad \Rightarrow \quad \vec{u} = \vec{v}_A + \vec{v}_B$$

So $\vec{v}_A$, $\vec{v}_B$ and $\vec{u}$ form a triangle, with $\vec{v}_A$ and $\vec{v}_B$ tip-to-tail.

2. The collision is elastic, so kinetic energy is conserved. Cancel $\frac{1}{2}m$:

$$u^2 = v_A^2 + v_B^2$$

3. That is Pythagoras' theorem for the triangle in step 1. A triangle whose sides obey Pythagoras has a right angle opposite the longest side u, so the angle between $\vec{v}_A$ and $\vec{v}_B$ is

$$\boxed{90^\circ}$$

The only exception is a head-on hit, where A stops dead and B takes all the velocity, so there is no angle to measure.

Real snooker balls roll and spin, so after the hit the cue ball's path bends away from the 90° line, but straight after contact the rule holds well.

Elastic and inelastic collisions

In a closed system **momentum is always conserved**. Kinetic energy is a different story. Total energy is always conserved too, but some of the kinetic energy can be transferred to other stores during the impact: the objects warm up slightly, they make a sound, and they may be permanently bent or dented.

DEFINITION

Elastic and inelastic collisions

In an **elastic** collision the total kinetic energy of the objects is the same before and after. In an **inelastic** collision the total kinetic energy after is less than before (momentum is still conserved).

How to decide

1. Use momentum conservation (in each direction) to find any unknown velocities.
2. Find the **speed** of every object before and after. In 2D, combine the components first with Pythagoras.
3. Work out $\frac{1}{2}mv^2$ (or $p^2/2m$) for every object and add them up, before and after.
4. Equal totals: elastic. Smaller total after: inelastic. If the objects stick together, it is certainly inelastic. A **bigger** total after means energy came from somewhere else, as in an explosion, where chemical or elastic potential energy is released.

COMMON MISTAKE

Kinetic energy is a **scalar**. Never resolve it into components or give it a minus sign for moving left. A ball moving left at 3 m s^{-1} has exactly the same kinetic energy as one moving right at 3 m s^{-1}, even though their momenta have opposite signs.

Collisions between hard steel ball bearings or snooker balls are very nearly elastic, and in kinetic theory collisions between gas molecules are taken to be perfectly elastic. A car crash, a ball of Plasticine hitting a wall, or a dart sticking into a dartboard are strongly inelastic.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Measuring "bounciness": the coefficient of restitution

Physicists and engineers give every pair of materials a number e, the coefficient of restitution:

$$e = \frac{\text{speed at which they separate}}{\text{speed at which they approach}}$$

measured along the line joining their centres. $e = 1$ is perfectly elastic and $e = 0$ means they don't bounce apart at all. Steel on steel is about 0.95 and a tennis ball on a court about 0.75. For a ball dropped onto the floor there's a neat trick: since $v = \sqrt{2gh}$ on the way down and up, $e = \sqrt{h_{\text{bounce}}/h_{\text{drop}}}$. A ball that bounces back to 64% of its drop height has $e = 0.8$. The collision simulation above uses e behind the scenes for its "Inelastic" setting.

Kinetic energy in terms of momentum

Collision problems are often easier if you can jump straight from momentum to kinetic energy without finding the velocity first. The link is a two-line derivation you need to be able to reproduce.

Deriving $E_k = p^2/2m$

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Start from the kinetic energy of a mass m moving at speed v:

$$E_k = \tfrac{1}{2}mv^2$$

2. Multiply the top and bottom by m. This changes nothing (it's multiplying by 1) but builds an mv on top:

$$E_k = \frac{m \cdot mv^2}{2m} = \frac{m^2v^2}{2m} = \frac{(mv)^2}{2m}$$

3. The bracket on top is the momentum, $p = mv$:

$$\boxed{E_k = \frac{p^2}{2m}}$$

Rearranged, $p = \sqrt{2mE_k}$. Use whichever direction the question needs.

Kinetic energy of a non-relativistic particle

FORMULA SHEET

$$E_k = \frac{p^2}{2m}$$

E_k: kinetic energy / J p: momentum / kg m s^{-1} m: mass / kg

"Non-relativistic" means much slower than light. Both $p = mv$ and $E_k = \frac{1}{2}mv^2$ start to go wrong as speeds approach 3.00×10^8 m s^{-1}, so this equation does too. For electrons accelerated through a few thousand volts it is fine.

KEY IDEA

Same momentum, different energy

When something at rest explodes into two pieces, they get **equal** sized momenta p. Then $E_k = p^2/2m$ says the lighter piece gets more kinetic energy, in the inverse ratio of the masses: $E_1/E_2 = m_2/m_1$. That's why a bullet carries almost all the energy and the gun's recoil is gentle, and why an alpha particle carries nearly all the energy released when a nucleus decays.

WORKED EXAMPLE

A stationary nucleus emits an alpha particle of mass 6.64×10^{-27} kg with kinetic energy 7.7×10^{-13} J. The nucleus left behind has mass 3.69×10^{-25} kg. Calculate its recoil kinetic energy and the fraction of the total energy carried by the alpha particle.

Show solution

Momentum of the alpha particle:

$$p = \sqrt{2mE_k} = \sqrt{2 \times 6.64 \times 10^{-27} \times 7.7 \times 10^{-13}} = 1.01 \times 10^{-19} \text{ kg m s}^{-1}$$

The total momentum was zero, so the nucleus recoils with the same size momentum in the opposite direction:

$$E_k = \frac{p^2}{2m} = \frac{(1.01 \times 10^{-19})^2}{2 \times 3.69 \times 10^{-25}} = 1.4 \times 10^{-14} \text{ J}$$

Fraction carried by the alpha particle:

$$\frac{7.7 \times 10^{-13}}{7.7 \times 10^{-13} + 1.4 \times 10^{-14}} = 0.98$$

Check with the mass ratio: $E_{\text{nucleus}}/E_\alpha = m_\alpha/m_{\text{nucleus}} = 0.018$, and $0.018 \times 7.7 \times 10^{-13} = 1.4 \times 10^{-14}$ J.

The same equation links to the de Broglie wavelength from Unit 2. An electron accelerated through 150 V gains $E_k = eV = 1.60 \times 10^{-19} \times 150 = 2.40 \times 10^{-17}$ J. Its momentum is $p = \sqrt{2mE_k} = \sqrt{2 \times 9.11 \times 10^{-31} \times 2.40 \times 10^{-17}} = 6.61 \times 10^{-24}$ kg m s^{-1}, so $\lambda = h/p = 6.63 \times 10^{-34}/6.61 \times 10^{-24} = 1.0 \times 10^{-10}$ m: about the spacing of atoms in a crystal, which is why electron beams diffract.

⚠ Beyond the spec · not part of the Edexcel IAL specification

The centre-of-mass frame, and the most energy a collision can lose

The **centre of mass** of two colliding objects moves at $v_{\text{cm}} = \dfrac{m_A v_A + m_B v_B}{m_A + m_B}$, which is the total momentum divided by the total mass. Momentum is conserved, so v_{cm} never changes, collision or not.

Now imagine running alongside at v_{cm}. In your frame the total momentum is zero: the objects rush at each other with equal and opposite momenta and leave with equal and opposite momenta. In an elastic collision, each one leaves with the same speed it arrived with, just in a new direction.

Back in the lab, the kinetic energy splits into two parts: the energy of the centre of mass moving, $\dfrac{p_{\text{total}}^2}{2(m_A + m_B)}$, plus the energy of the motion relative to it. The first part is fixed by momentum conservation, so it can **never** be lost. Only the second part can be lost, and all of it goes when the objects stick together. So "stick together" is the most inelastic a collision can be.

Core Practical 10: analysing collisions with ICT

CORE PRACTICAL

Use ICT to analyse collisions between small spheres, e.g. ball bearings on a table top

Ball A is rolled down a short ramp so that it strikes ball B, which is at rest on a flat, level table, slightly off-centre. A camera (a phone on a stand is fine) films the collision from **directly above**, with a metre rule or grid in the same plane as the balls for scale.

1. Measure the mass of each ball with a top-pan balance.
2. Load the video into analysis software (such as Tracker). Set the scale using the ruler, set the origin and axes, and mark the centre of each ball frame by frame.

3. The software plots x and y against t for each ball. The **gradient** of each straight section gives v_x and v_y before and after the collision.
4. Calculate the total p_x and total p_y before and after. They should agree within experimental uncertainty.
5. Calculate the total kinetic energy before and after to decide whether the collision was elastic.

- **Camera not directly overhead** distorts distances (parallax). Use a spirit level on the camera and keep the scale rule in the plane of the balls.
- **Motion blur** makes centres hard to mark. Use bright lighting and a high frame rate.
- Friction slowly slows the balls, so use frames close to the collision, and fit a line through several points rather than using just two.
- Rolling balls also have rotational kinetic energy, so the "missing" energy you calculate is not all lost in the impact.

Quick checks

QUICK CHECK

A 0.50 kg ball hits a wall at 4.0 m s^{-1} and bounces straight back at 4.0 m s^{-1}. What is the magnitude of its change in momentum?

- 0 N s
- 2.0 N s
- 4.0 N s
- 8.0 N s

The speed is unchanged but the velocity reverses: $\Delta p = 0.50 \times (-4.0 - 4.0) = -4.0$ N s. Zero is the trap: momentum is a vector, so a change of direction is a change of momentum.

QUICK CHECK

Why does an airbag reduce the force on a driver's head in a crash?

- It reduces the head's change in momentum.
- It increases the time taken for the head to stop, so a smaller force gives the same impulse.
- It absorbs the head's momentum so momentum is not conserved.
- It reduces the impulse on the head.

The head goes from moving to stopped either way, so Δp (and the impulse) is the same. $F = \Delta p / \Delta t$, so a longer Δt means a smaller average force.

QUICK CHECK

Puck A moves along the x-axis and hits stationary puck B. Afterwards B moves off at 30° below the x-axis. Which statement about A after the collision must be true?

- A moves along the x-axis.
- A has a component of velocity above the x-axis (or A is at rest).
- A moves at 30° below the x-axis.
- A must stop.

Before the collision the total y-momentum is zero, so afterwards A's y-momentum must cancel B's downward component. The only way A could avoid this is by having no momentum at all.

QUICK CHECK

Particles X and Y have the same momentum. X has four times the mass of Y. What is the ratio (kinetic energy of X) : (kinetic energy of Y)?

- 1 : 4
- 4 : 1
- 1 : 2
- 1 : 16

$E_k = p^2/2m$. With the same p, kinetic energy is inversely proportional to mass, so the heavier particle has a quarter of the energy.

QUICK CHECK

Two trolleys collide and stick together. No external forces act. Which statement is correct?

- Total kinetic energy and total momentum are both conserved.
- Total kinetic energy is conserved but total momentum is not.
- Total momentum is conserved but total kinetic energy is not.
- Neither is conserved, but total energy is.

With no external force, momentum is always conserved. Objects sticking together is always inelastic, so some kinetic energy is transferred to thermal energy and sound. Total energy is conserved too, which is why the last option is a trap: momentum is conserved as well.

QUICK CHECK

Which of these is not a correct unit for impulse?

- N s
- kg m s^{-1}
- N s^{-1}
- J s m^{-1}

Impulse is force × time, so N s, which equals kg m s^{-2} × s = kg m s^{-1}. Since 1 J = 1 N m, J s m^{-1} is also N s. N s^{-1} would be a rate of change of force.